

BTECH/CSE/IT/ECE/EE/EIE/CE/ME/Even/Sem-II/CS201/2020-2021
PAPER TYPE: REG+SUP+RESUP
YEAR: 2021
REGULATION: 2018

PROGRAMMING FOR PROBLEM SOLVING

CS201

TIME ALLOTTED: 3 HOURS

FULL MARKS: 70

The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable

GROUP – A (Multiple Choice Type Questions)

1. Answer any **ten** from the following, choosing the correct alternative of each question: **10×1=10**

SL. NO.	Question	CO No.	Marks
(i)	The smallest integer that can be represented using 8-bits in signed 2's complement system is a) -256 b) -128 c) -127 d) 0	1	1
(ii)	(1217) ₈ is equivalent to a) (1217) ₁₆ b) (028F) ₁₆ c) (2297) ₁₀ d) (0B17) ₁₆	1	1
(iii)	The declaration float *a[5]; is a) a normal pointer b) an ordinary array c) an array of pointer d) a pointer to an array	3	1
(iv)	Static memory allocation allocates memory during time. a) Compile b) Run c) Load d) Link	4	1
(v)	If a variable is a pointer to a structure, then which of the following operator is used to access data members of the structure through the pointer variable? a) . b) & c) * d) ->	3	1
(vi)	Which of the following is a bitwise operator? a) < b) >= c) << d) &&	1	1
(vii)	What is the default return type of main() function in C programming? a) void b) int c) unsigned int d) NULL	3	1

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(viii)	Which of the following can be used to represent signed number? a) Sign Magnitude System b) Signed 1's Complement System c) Signed 2's Complement System d) All of the Above	1	1
(ix)	char str[100]="Hello World!"; How many bytes are allocated by the above declaration? a) 101 b) 100 c) 13 d) 12	3	1
(x)	Which of the following operator(s) in C Programming Language can be considered as unary operator? a) + b) - c) ~ d) All of the above	1	1
(xi)	What is default storage class of variables in C language? a) auto b) local c) extern d) global	3	1
(xii)	If a variable is a pointer to a structure, then which of the following operator is used to access data members of the structure through the pointer variable? a) . b) & c) * d) ->	3	1

GROUP – B*
(Short Answer Type Questions)

Answer any *three* from the following: 3×5=15

SL. NO.		CO No.	Marks
2.	Define a macro to find the smallest number among 3 given numbers. Call it from main function to print the smallest number.	3	5
3.	(a) What is a logical expression in C? Give example.	1,2,3	1+1
	(b) Evaluate the following expression (value of X). Show all the steps. X= 6-3<<2	1,2,3	3
4.	(a) What is the return type of calloc function? What is the difference between calloc and malloc function?	4	1+2
	(b) Create an 2D float array of r rows and c columns dynamically.	3,4	2
5.	(a) What is unary operator in C? Give two examples of unary operators in C.	1	1+1
	(b) Evaluate the following expression (value of X). Show all the steps. X= - ~ 11	1	3
6.	(a) Write the general syntax of for loop.	1	2
	(b) Write a flow chart that correspond to the following for loop for (p=0; p<n; ++p) statement1; statement2;	1	3

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GROUP – C*
(Long Answer Type Questions)

Answer any **three** from the following: **3×15=45**

SL. NO.		CO No.	Marks
7.	(a) Write a C program to check whether a given string is a sub-string of another string or not. Both the strings are input from user. Do not include string.h header.	3	10
	(b) Write a C program to check whether a given integer is an even number or not using bit operation.	2,3	5
8.	(a) Explain file and FILE in C language	3	2+5
	(b) Write C code to open a file named input.txt for reading and writing (write only the line) with proper error checking. Name of the respective file should be taken from command line.	2,3	3+5
9.	(a) Explain the return type printf and scanf return in C?	1	2+2
	(b) What are the different types storage class in C?	3	4
	(c) What is scope and lifetime of a different types of variables in C with proper example.	3	7
10.	(a) Given an unsorted array A of size N of non-negative integers, find a continuous sub-array which adds to the given number (taken as input). Print the starting and end index of the sub-array if exists, otherwise print -1.	2,3	10
	(b) Write a C program to check whether a number taken as input is prime or not.	2	5
11.	(a) “Passing the base address of an array than the entire array is advantageous” – Justify this statement with example.	3	5
	(b) “Macro constant should be used for providing size of an array during declaration.” – Justify this statement with example.	3	5
	(c) Critically comment on initializing union variable at the place of declaration with respect to structure variables.	3	5